Validation Flight

APPN Aerial SOP — Locked revision 1.0

Validation Flight Procedure

Version 0.1 – DRAFT, April 2026

> **Important**
>
> This protocol outlines the **validation flight procedures** used to periodically check that APPN aerial payloads (GOBI, CALViS) remain within spectral and spatial calibration, and to support the APPN APEx experimental program. It defines three validation flight types, each with its own purpose, frequency, and acceptance criteria.
>
> Validation flights are intended to be flown:
>
> - At **regular intervals** by every node, so that data products remain comparable across nodes and over time.
> - **Any time a sensor is returned to service** after repairs, firmware updates, or other maintenance.
> - **Whenever a flight parameter is being formally tested** under the APEx program.
>
> For routine operational surveys, see the Standard Flight Procedure.
>
> For sensor-specific equipment, procedures, and safety information, see the relevant sensor fieldbook:
>
> - CALViS Fieldbook
> - GOBI M350 Fieldbook
> - GOBI IF1200 Fieldbook

> **Caution**
>
> **This document is being completely restructured.** Section ordering, scope, and procedural detail will all change in upcoming revisions. Treat the material below as a placeholder scaffold only and do not rely on the current content for field use until this notice is removed.

> **Important**
>
> **Document status — work in progress.** This protocol is a draft and requires further revision before it can be considered final. New figures also need to be produced to illustrate the procedures described below.
>
> **Outstanding TODOs:**
>
> **Content review**
>
> - [ ] Define and confirm the **frequency** of each validation flight type (currently `_TODO_`).
> - [ ] Confirm the **acceptance criteria / pass-fail thresholds** for each validation flight type with the Field EWG.
> - [ ] Cross-check terminology and links against the sensor fieldbooks and the Standard Flight Procedure.
> - [ ] Confirm equipment checklists with field operators (all three validation flight types).
>
> **Figures to produce** (see master list in `/IMAGE_TODO.md`)

- ☐ **Diagram** — Spectral validation flight: panel layout, flight-line orientation, and GCP distribution.
- ☐ **Diagram** — Spatial validation flight: GCP layout (including elevated / paired GCPs), flight-line geometry, and any check targets.
- ☐ **Diagram** — APEx experimental flight: reference layout used as the starting point for parameter sweeps.
- ☐ **Photo (suggested)** — Example validation site with uniform crop cover, fixed repeatable location, clear of tall obstructions (see Site selection).
- ☐ **Chart (suggested)** — Sample QC plot showing spectral validation metrics (e.g. panel-derived reflectance drift, band-to-band offsets) annotated as pass vs fail (see Acceptance criteria).

Cross-links to add

- ☐ Processing pipeline page describing how validation outputs are analysed (radiometric drift, geometric error reports).
- ☐ QA process page describing how validation results feed into the node-level QA record.

Pending APEx decisions (revise before season start)

- ☐ Lock the standard validation flight footprint (area, altitude, speed, overlap) once APEx parameter sweeps complete.
- ☐ Confirm minimum solar elevation / time-of-day window for routine validation flights.
- ☐ Confirm whether spectral and spatial validation can be combined into a single overflight or must be flown separately.

Document Structure

This protocol is organised into two parts:

1. **Background and common elements of all validation flights** — shared setup rules that apply to every validation flight design:

- Site selection
- Flight-line orientation and timing
- Panel setup
- Ground Control Points
- Records and metadata

2. **The three validation flight types**, each with its own overview, when-to-use criteria, flight-design notes, and equipment list:

- Spectral validation flight — periodic check that the radiometric / spectral response of the sensor remains within tolerance and is comparable across nodes.
- Spatial validation flight — periodic check on geometric accuracy (horizontal, vertical, and across-track / between-flight-line registration).
- APEx experimental flight — parameter-sweep flights used by the APPN APEx program to test and refine flight settings, sensor configuration, and processing options.

Background and common elements of all validation flights

Important

This section describes the **placement of GCPs and reflectance panels, site selection, and metadata recording** that is common to all validation flight types described below.

All placements given here (distances, positions, panel locations, GCP distributions) are **approximate guidelines, not strict requirements**. Operators should adapt the layout to the realities of the site:

- Use the **nearest existing path, bare strip, or tramline** to access and place panels and GCPs.
- **Do not place panels or GCPs inside experimental plots** or anywhere they would damage the crop,

disturb a treatment, or obstruct other field operations.
- Keep panels and GCPs clear of shadows (from trees, infrastructure, the operator, or the UAV itself) and away from the take-off / landing zone.

Caution

Keep panels and GCPs away from the edge of the capture window.

Depending on the exact flight parameters (altitude, overlap, turn geometry), GOBI and CALViS can fail to capture data in the outermost portion of the planned capture polygon. Place all validation targets inside an **effective capture area** set back from each edge of the planned polygon.

See the Standard Flight Procedure background section for the full rationale and the recommended ~10% inset.

Site selection

Validation flights can be flown over any safe and convenient location, but should ideally be flown:

- Over a **closed crop canopy or uniform grass cover** wherever possible, to provide a stable spectral and structural reference.
- At a **fixed, repeatable site at each node**, so that successive validation flights can be compared directly without site effects confounding the result.
- Clear of nearby tall obstructions (trees, sheds, powerlines) that would cast shadows or restrict flight-line geometry.

  TODO: Add guidance on minimum site footprint, and on documenting the chosen validation site at each node.

Note

Image needed (photo, suggested): Example field photograph of an appropriate validation site — uniform crop or grass cover, fixed repeatable location at the node, clear of tall obstructions — to anchor the site-selection criteria above.

Flight-line orientation and timing

- Default flight-line orientation is **North–South**, flown as close as practical to **solar noon**.
- Record the **actual flight-line orientation, start time, and end time** for every validation flight. Validation flights are only comparable across nodes and over time if these parameters are consistent — and where they cannot be, they must be logged.

  TODO: Confirm minimum solar elevation, maximum permissible wind speed, and any other go / no-go weather criteria for validation flights, once APEx results are available.

Panel setup

The panel setup for validation flights follows the same rules as for operational surveys — see Panel Setup and Validation Panel Setup in the Standard Flight Procedure for the full procedure (table, levelling, ordering, orientation, height recording, paired GCP).

The key validation-specific points are:

- Validation flights **always include the GRYFN 2-panel validation set** with its paired elevated + ground GCPs.
- Where two ELM panel sets are available, both should be deployed (see the Dual ELM panel flight layout).
- Panel positions and the **panel surface height above the ground** must be recorded for every validation flight.

Ground Control Points

Validation flights use the same minimum **5-GCP layout** described in the Standard Flight Procedure (Ground Control Points), with **at least one GCP placed inside each flight line** and the paired elevated + ground GCPs at the validation panel table.

Additional GCPs are encouraged for validation flights, since the purpose of the flight is precisely to measure spatial accuracy.

Records and metadata

For every validation flight, record (at minimum):

- **Flight identifier**, node, operator, date, and start / end time.
- **Sensor and platform** (including firmware / software versions).
- **Site identifier** and any deviations from the standard site.
- **Flight parameters** actually flown (altitude, speed, overlap, flight-line angle).
- **Weather**: cloud cover, wind speed and direction, air temperature, approximate solar elevation.
- **Panel configuration**: number of panel sets, table heights, position relative to flight lines.
- **GCP layout**: positions and any deviations from the standard 5-GCP layout.
- **Reason for the flight**: scheduled interval, post-maintenance check, APEx experiment, etc.

  TODO: Link this to a standard validation flight log template / wiki page once available.

Spectral validation flight

Overview

The spectral validation flight is a periodic check that the **radiometric and spectral response** of the GOBI and CALViS payloads remains within tolerance — both against itself over time, and against other nodes flying the equivalent payload.

Rationale:

- **Drift detection.** Detects gradual changes in sensor response (e.g. due to ageing optics, panel degradation, or seasonal illumination differences) before they affect operational data quality.
- **Inter-node comparability.** Provides a common reference flight pattern that lets every node check that its calibrated reflectance products agree with those of other nodes flying the same payload.
- **Post-maintenance check.** Confirms that a sensor returned to service after repair, replacement, or firmware change still produces reflectance values consistent with its pre-maintenance baseline.

When to use this flight:

- At **regular intervals** by every node — *TODO: define interval (e.g. monthly during the field season, quarterly out of season).*
- **Any time a sensor is returned to service** after repairs, firmware updates, or replacement of optical components.
- Whenever the **standard radiometric panels** are replaced, re-characterised, or otherwise changed.

Flight design

> **Note**
>
> **Image needed (diagram):** Top-down annotated diagram of the spectral validation flight layout — panel placement (dual ELM where available), validation panel position, flight-line orientation and overlap, and the 5-GCP layout.

TODO: Insert annotated diagram showing the spectral validation flight layout — panel placement (dual ELM where available), validation panel position, flight-line orientation and overlap, and the 5-GCP layout.

The current draft layout is the dual-panel flight described in the Standard Flight Procedure, flown over the node's fixed validation site. Specifically:

- **Survey footprint:** approximately **7,200 m² (60 m × 120 m)** rectangle (working draft — to be locked once APEx parameter sweeps complete).
- **Flight lines:** 5 lines, oriented North–South by default, flown close to solar noon.
- **Panels:** GRYFN 4-panel ELM sets at ~30% along flight line 1 and ~30% along flight line 4 (the second set captured under a different flight-line direction relative to the first); GRYFN 2-panel validation set near the centre of flight line 3.
- **GCPs:** one GCP per flight line in a diagonal pattern at 10%, 30%, 50%, 70%, and 90% of the line, plus the paired elevated / ground GCP at the validation panel table.

> **Note**
>
> **Image needed (diagram):** The figure above is a **legacy placeholder** retained only so the section is not blank. Replace with a new annotated top-down diagram of the spectral validation flight — dual ELM panel placements, validation panel position, N–S flight-line orientation, edge buffer, and the 5-GCP layout, plus a legend for GCPs, GRYFN reflectance panel sets, and the independent validation panel. Tracked in IMAGE_-TODO.md.

IF1200 + CALViS or GOBI flight parameters *(draft)*

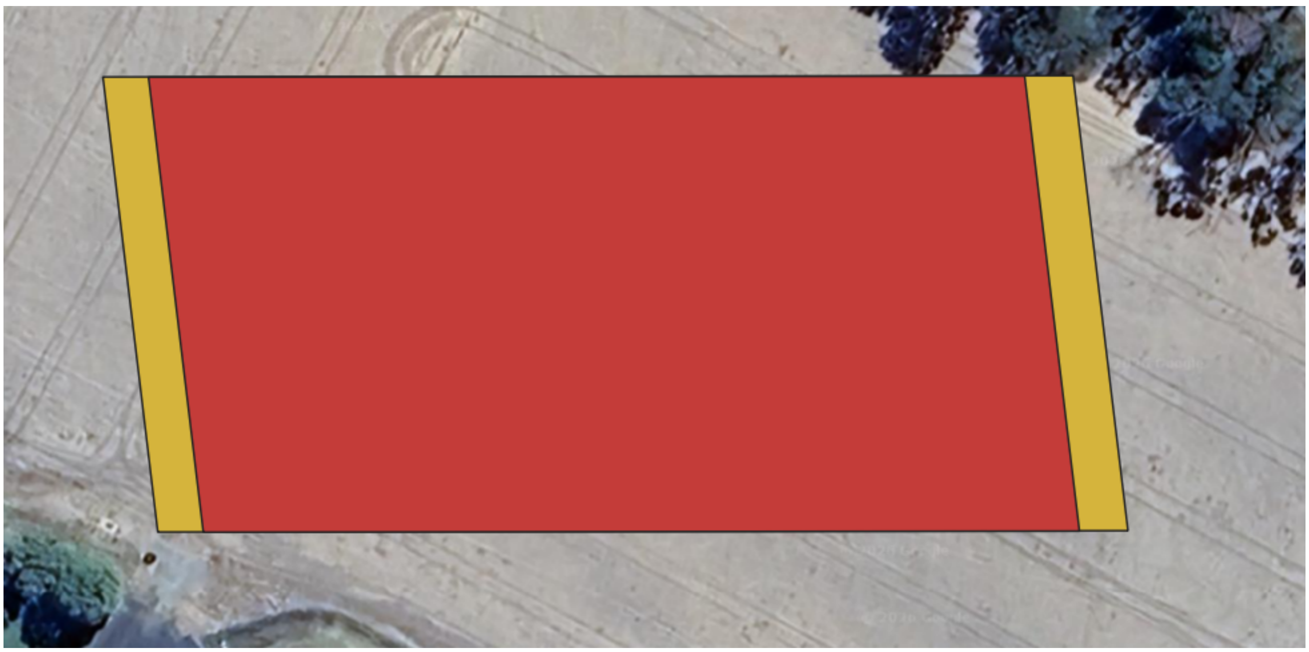
1. Using QGroundControl, set up a new flight mission at the node's fixed validation site.
2. Create a rectangular survey area of approximately **6,200 m²**.
3. Export the polygon and import it into the *HPI Polygon Tool*.
4. Use the parameters below for flight planning in QGroundControl:

Parameter	Setting
Altitude	50 m
Flight speed	3.2 m/s
Trigger Distance	Leave as default (number is irrelevant)
Spacing	12 m
Angle	180°
Turnaround Distance	10 m

4. Load this mission onto the IF1200.
5. **TODO: Insert exposure-setting procedure for the spectral validation flight.**

DJI M350 RTK + GOBI flight parameters *(draft)*

1. Using DJI Pilot 2, set up a new flight mission at the node's fixed validation site.
2. Create a rectangular survey area of approximately **6,200 m²**.
3. Create a smaller rectangular *capture* polygon inside the *survey* polygon, inset by approximately **2 × the flight speed (~7 m)** in the direction of flight. The figure below outlines this — the red polygon is the *capture* polygon, the yellow is the *survey* polygon.



1. Export the polygon and import it into the HPI Polygon Tool.

2. Use the parameters below for flight planning in DJI Pilot 2:

Parameter	Setting
Altitude	50 m
Flight speed	3.2 m/s
Side Overlap	49%
Frontal Overlap Ratio	80%
Course Angle	180°
Margin	0
Elevation Optimisation	FALSE

5. **TODO: Insert exposure-setting procedure for the spectral validation flight.**

Acceptance criteria

> **Note**
>
> **Image needed (chart, suggested):** Sample QC chart (e.g. exported plot) showing spectral validation metrics over time — panel-derived reflectance drift, band-to-band offsets, spectroradiometer–UAV mismatch

> — with the pass/fail thresholds annotated.

TODO: Define quantitative pass / fail thresholds for the spectral validation flight (e.g. maximum drift in panel-derived reflectance, maximum band-to-band offset, maximum spectroradiometer–UAV mismatch).

Ground Equipment Required

- ☐ UAV, controller, and batteries
- ☐ GOBI or CALViS payload
- ☐ 2 × GRYFN 4-panel ELM set (11%, 30%, 56%, 82%) — if only 1 is available, fall back to the single-panel layout
- ☐ 1 × GRYFN 2-panel validation set (20%, 45%)
- ☐ 5 × Propeller Aeropoints or equivalent GCPs
- ☐ 2 × Folding tables (1 per panel set)
- ☐ Spirit level (to level the tables)
- ☐ Tape measure
- ☐ Optional spectroradiometer (SVC, FieldSpec) for an independent spectral reference
- ☐ Downwelling radiation sensor *(TODO: add details)*
- ☐ Weather station *(TODO: add details)*

> **Important**
>
> All other equipment listed in the relevant sensor fieldbook is also required.

Spatial validation flight

Overview

The spatial validation flight is a periodic check on the **geometric accuracy** of the GOBI and CALViS payloads — covering horizontal position, vertical / elevation accuracy, and across-track or between-flight-line registration.

Rationale:

- **Detects geometric drift.** Picks up changes in IMU / GNSS / boresight alignment that would otherwise show up only as subtle stitching errors in operational products.
- **Validates the standard 5-GCP layout.** Confirms that the GCP configuration recommended in the Standard Flight Procedure is actually resolving the spatial errors it is intended to detect.
- **Post-maintenance check.** Confirms that a sensor returned to service after physical disassembly, mount changes, or boresight recalibration still produces geometrically consistent products.

When to use this flight:

- At **regular intervals** by every node — *TODO: define interval.*
- **Any time** a sensor or its mount has been physically disturbed (e.g. boresight recalibration, lens / optics swap, IMU replacement, remount on a different airframe).
- After any **firmware or processing-software update** that could affect georeferencing.

Flight design

> **Note**
>
> **Image needed (diagram):** Top-down annotated diagram of the spatial validation flight layout — expanded GCP distribution, paired elevated / ground GCPs, and any additional check targets used to characterise across-track and vertical error.

TODO: Insert annotated diagram showing the spatial validation flight layout — extended GCP distribution, paired elevated / ground GCPs, and any additional check targets used to characterise across-track and vertical error.

Working draft layout:

- Same survey footprint and flight-line geometry as the Spectral validation flight (so that the two validations can be flown back-to-back at the same site if scheduling allows).
- **Expanded GCP layout:** the standard 5-GCP layout (one per flight line in a diagonal pattern) **plus** additional GCPs near the corners and at off-axis positions between flight lines, to give redundant checks on each spatial-error mode described in the Standard Flight GCP rationale.
- **Paired elevated + ground GCPs** at the validation panel table to check vertical accuracy and across-track tearing.
- Optional **independently surveyed check targets** (e.g. permanent ground markers with known coordinates) to provide an external reference independent of the flight's own GCPs.

Acceptance criteria

> *TODO: Define quantitative pass / fail thresholds for the spatial validation flight (e.g. RMSE in horizontal position, vertical offset between elevated and ground GCPs vs measured table height, across-track tearing between adjacent flight lines).*

Ground Equipment Required

- ☐ UAV, controller, and batteries
- ☐ GOBI or CALViS payload
- ☐ 1 × GRYFN 4-panel ELM set (for radiometric reference, even though spectral accuracy is not the primary target)
- ☐ 1 × GRYFN 2-panel validation set (for the paired elevated / ground GCP arrangement)
- ☐ **Extended GCP set** — at least 5 (the standard layout) plus additional GCPs for corners and off-axis positions; more is better
- ☐ 2 × Folding tables (1 per panel set)
- ☐ Spirit level (to level the tables)
- ☐ Tape measure
- ☐ Optional independently surveyed check targets

> **Important**
>
> All other equipment listed in the relevant sensor fieldbook is also required.

APEx experimental flight

Overview

APEx experimental flights are **parameter-sweep flights** used by the APPN APEx program to test and refine flight settings, sensor configuration, and processing options. Unlike the spectral and spatial validation flights — which are flown under fixed parameters to detect change — APEx flights deliberately vary one or more parameters at a time and compare the resulting products.

Rationale:

- **Parameter discovery.** Provides the empirical basis for choosing the operational defaults locked into the Standard Flight Procedure.
- **Site- and platform-specific tuning.** Allows nodes to test variations relevant to their local conditions (illumination range, canopy types, terrain, airframe).
- **Documented evidence base.** Generates the data behind APPN-wide recommendations on flight altitude, speed, overlap, exposure, flight-line orientation, etc.

When to use this flight:

- Whenever a flight parameter is being **formally tested** under the APEx program.
- During **APPN Calibration Week** or equivalent campaigns where multiple parameter sweeps are scheduled together.

- Any time a node wants to **propose a change to the Standard Flight Procedure** — the change must be supported by an APEx flight comparing the proposed setting against the current default.

Flight design

> **Note**
>
> **Image needed (diagram):** Top-down annotated diagram of the APEx reference layout — the spectral validation flight footprint with standard panel and GCP positions clearly marked as the baseline against which parameter sweeps are compared.

APEx flights start from the Spectral validation flight layout (panels, GCPs, site, base flight parameters) and then vary the parameter under test. The unchanged baseline flight is **always flown as a control alongside the experimental variant(s)**, so that the effect of the parameter change can be isolated from day-to-day illumination and weather variability.

For each APEx flight, record:

- The **parameter being varied** and the values tested.
- The **control / baseline parameters** used for comparison.
- The **flight order** (control vs experimental) and elapsed time between flights.
- All standard validation-flight metadata (see Records and metadata above).

APEx test catalogue ***(draft)***

Stoplight rating (used during APEx planning):

- **Green** — go for Calibration Week
- **Blue** — yes at Calibration Week if achievable, but follow up with tests later in the season
- **Red** — abandon for now

Calibration Week

1. **Flight height** (determined by GSD required)
2. 20 m to 120 m
3. Mixed pixels
4. **Flight speed, frame period, front oversampling, and gain settings**
5. **Exposure setting protocol**
6. Set the exposure with a fixed angle / bracket?
7. Setting at exposure with lower reflectance panel (50–60%)?
8. Three-point ELM vs four-point ELM
9. **Side overlap settings** (tearing)
10. **Time of day (solar angles)**
11. Angles between 10° (Adelaide 8 am winter solstice) and 85° (Gatton solar noon summer solstice)
12. Ideally every 5° or 10°
13. Covers time of day and seasonality

Post Calibration Week

1. **UWA-DPIRD?** Crosstrack illumination — April
2. Panels set up in the middle and edge of the flight lines
3. Very high flights seem to be more pronounced
4. **UQ-CSU?** Flight orientation (should be done in fully closed crop canopy)
5. Default is N–S
6. E–W
7. NE–SW
8. NW–SE

Test different **wind speed and climate conditions** (may not be a discrete experiment if a high-frequency weather station is set up at every location and the data are mined post-season):

1. Minimum flight speed may change in different conditions.
2. There may be a maximum permissible wind speed for these sensors and UAV platforms.
3. Movement of plants and mixing of pixels.
4. Whether flights are acceptable under uniform cloud cover.

Acceptance criteria

APEx flights do not have pass / fail criteria in the same sense as the spectral and spatial validation flights — their purpose is **measurement, not certification**. Each APEx experiment must nevertheless define, in advance:

- The **metric(s)** that will be used to compare the experimental variants against the control.
- The **threshold(s)** at which the experimental setting would be recommended over the current default.
- How the result will be **reported back** to the Field EWG and, if adopted, propagated into the Standard Flight Procedure.

Ground Equipment Required

- ☐ All equipment listed for the Spectral validation flight above
- ☐ **Any additional equipment specific to the experiment** (e.g. extra panel sets for cross-track illumination tests, independent weather station, additional GCPs for orientation sweeps, spectroradiometer for ELM comparisons).

> **Important**
>
> All other equipment listed in the relevant sensor fieldbook is also required.
